

Project Title: AI-Powered Career and Skill-Gap Recommendation System Using Canadian Job Bank Data, O*NET Occupational Information, and CIP Educational Pathways

Course: Capstone Project (DAMO-699-1)

Program: Master of Data Analytics, University of Niagara Falls Canada

Instructor: Mehdi Mostofi

Group Members:

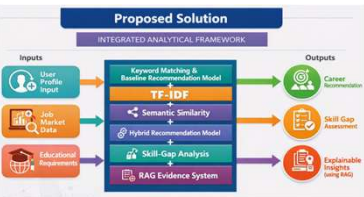
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Project Overview

This project develops an integrated career recommendation and skill-gap analysis system that connects candidate profiles with suitable occupations using NLP, semantic matching, occupational skill requirements, Canadian labour-market demand and educational pathways.



Background



Research Questions

Primary Research Question

RQ1: How much better can an integrated career recommendation system, using Canadian Job Bank labour-demand data, O*NET occupational information, and CIP educational pathways, improve career and skill-gap recommendations compared to traditional keyword-based matching?

Secondary Research Questions

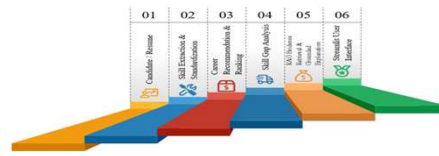
- RQ2: Which selected occupations have the most postings and vacancies in Canada?
- RQ3: How does the demand for selected occupations differ across provinces or territories?
- RQ4: Which O*NET knowledge areas, skills, and abilities are most important for selected positions?
- RQ5: What educational programs are associated with the selected occupations based on the CIP-SOC crosswalk?
- RQ6: How accurately can the system identify skills, education, experience, certifications and projects from the resumes of the selected applicants?

Data & Data Quality

Data Sources, Preprocessing and Quality Assurance



System Architecture



Methodology

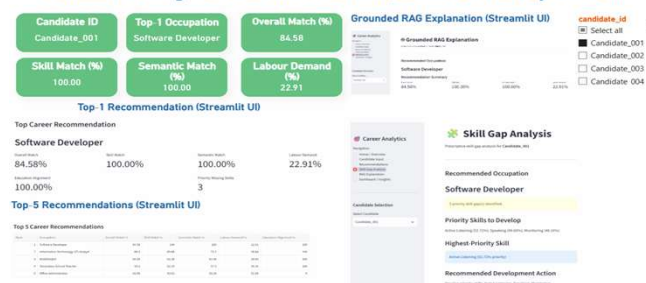


Deployment & Analytics Products

Analytical Approach & Model Diagnostics



Candidate Insights: Recommendations, Skills & RAG Explanation



Model Evaluation & RAG Assessment



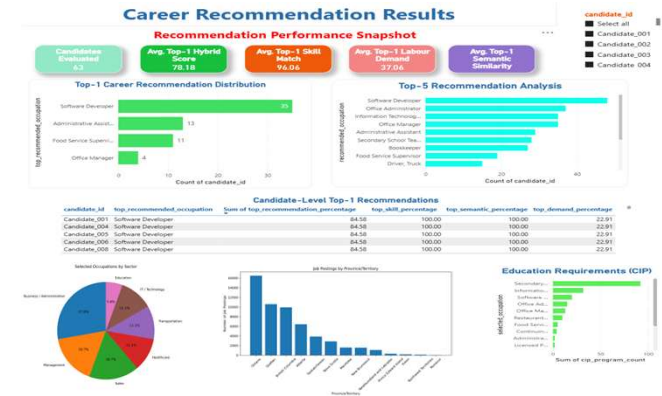
RAG Architecture



Example Candidate RAG Evidence Package

Evidence Component	Example Output
Candidate	Candidate_001
Recommended occupation	Software Developer
Hybrid score	84.58%
Priority skill gaps	Active Listening, Monitoring, Speaking
O*NET evidence	RAG document IDs
Task evidence	Occupational task documents
Labour-market evidence	Canadian demand document
Education evidence	CIP pathway document

Key Findings & Visualizations



- Strong alignment** - Most candidates show very high skill and semantic match (often 100%), meaning their resumes closely fit the occupations recommended by the model.
- Software Developer dominates** - Over half of all candidates receive "Software Developer" as their top recommendation, showing strong transferable digital skills across the group.
- Labour demand varies** - Even when skill match is high, labour-market demand scores differ widely, meaning some jobs are a great personal fit but not always in high demand.
- Top-5 recommendations diversify** - When looking beyond the top-1 result, candidates match with a broader set of occupations like IT Analyst, Bookkeeper, Office Administrator, and Teacher.

User Interface

Home Page



Live Result



Scan Me



Limitations

- Resume/skill extraction errors
- NOC-SOC-O*NET mapping limitations
- Labour market demand changes over time
- Relatively small candidate dataset (63 candidates)
- Limited labelled data for supervised Machine Learning
- Semantic-model limitations

Future Research

- Expand the candidate dataset to improve generalizability.
- Introduce larger labelled outcome datasets to support supervised learning and outcome-based validation.
- Improve semantic recommendation models to strengthen contextual matching.
- Integrate real-time labour-market APIs for continuously updated demand information.
- Enhance and scale the deployed Streamlit application for broader practical use.

Strategic Insights

- Candidates**
 - Prioritize high-importance missing skills to improve career readiness.
- Universities**
 - Align academic programs with occupational skill requirements.
- Employers**
 - Use skill-gap evidence to identify workforce development needs.
- Career Services**
 - Provide personalized occupation and skill-development pathways.

